CNENGR202 Instrumentation and Control

Part 2 – L7 Signal conditioning

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Signal conditioning

- Signal conditioning: Manipulating an analogue signal thus it meets the requirements of the next stage for further processing.

- Can roughly split the three types of instrumentation required as thus:
  - Conversion to a required format
  - Signal amplification
  - Signal filtering from noise

Convert a charge into a useable voltage signal: Charge amplifier

- When a semi-conductor detector is used to measure X-rays, the output signal can be weak charge with a pulse width of several tens of nanoseconds. Op_amp integrators are often used, with high impedance they integrate weak charge pulses, converting them into voltages.

C_t: Transducer capacitance

C_c: Cable capacitance

C_f: Feedback capacitor

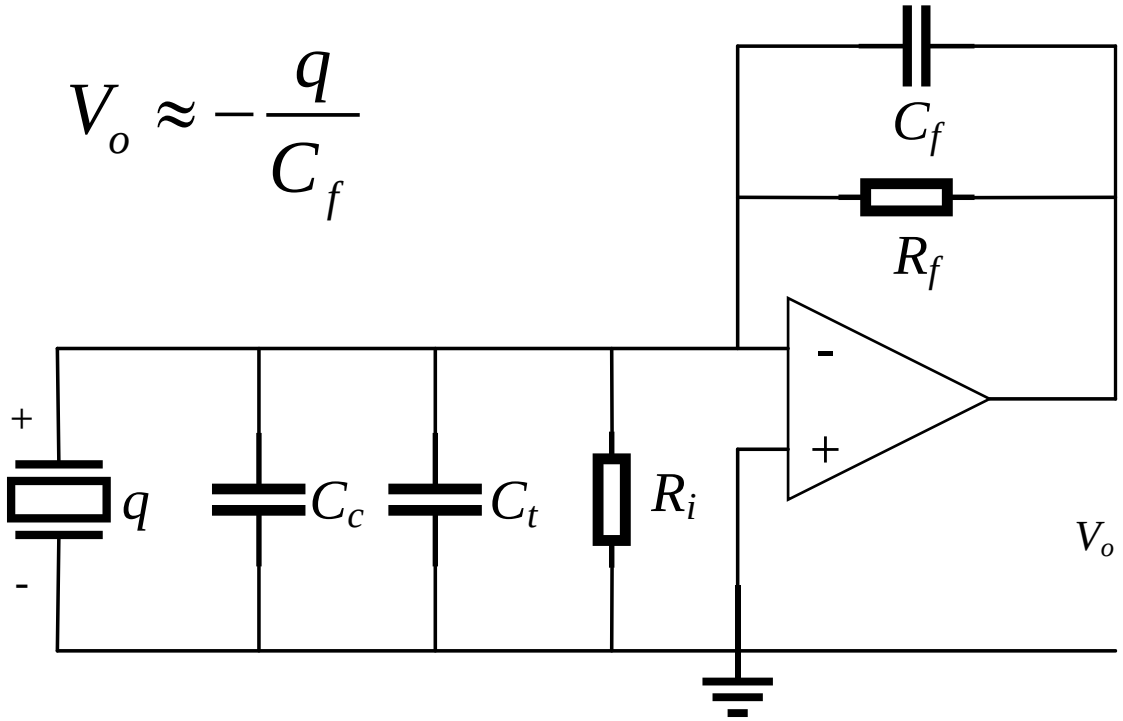
R_f: Time constant resistor

R_i: Insulation resistance (input)

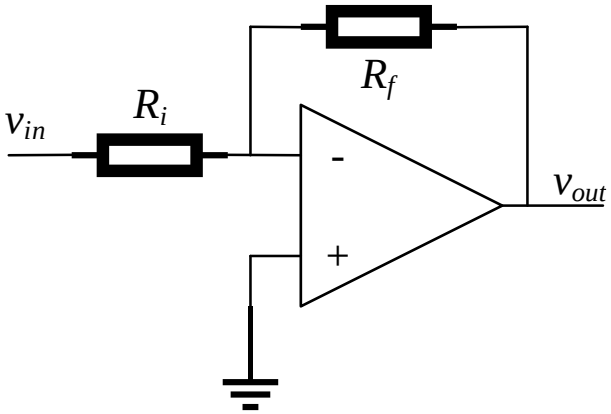
q :Transducer charge

V_o: Output voltage

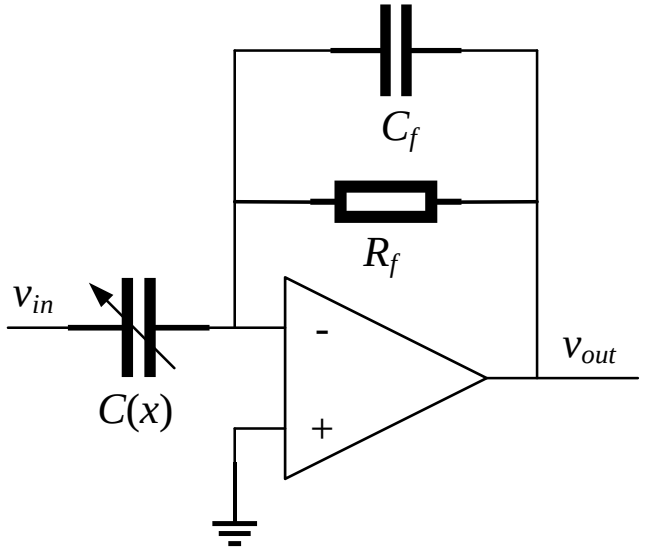
$$V_o \approx -\frac{q}{C_f}$$



Operational Amplifier - 1

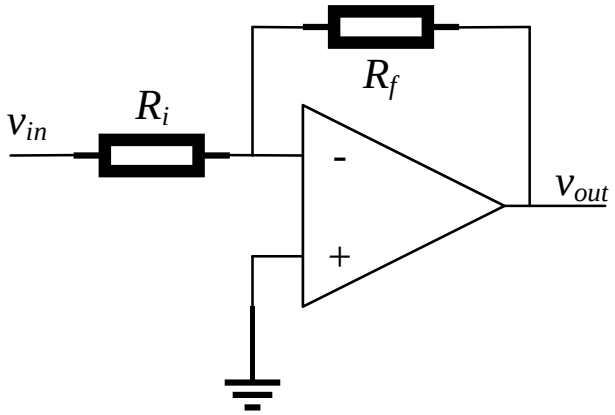


$v_{out} = ?$

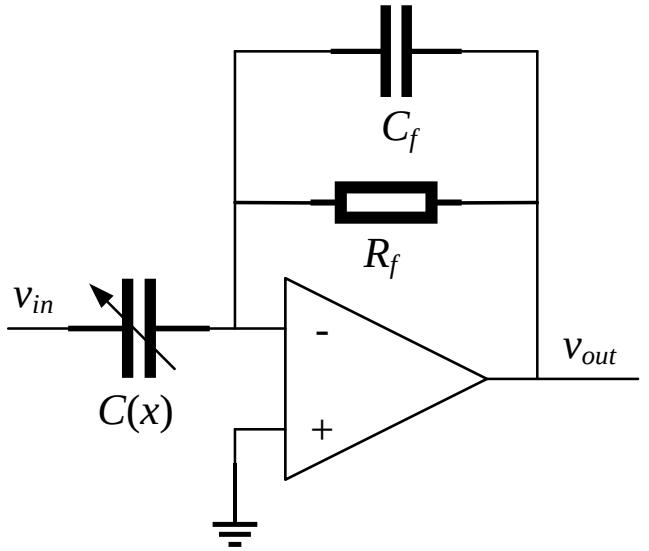


$v_{out} = ?$

Operational Amplifier - 2

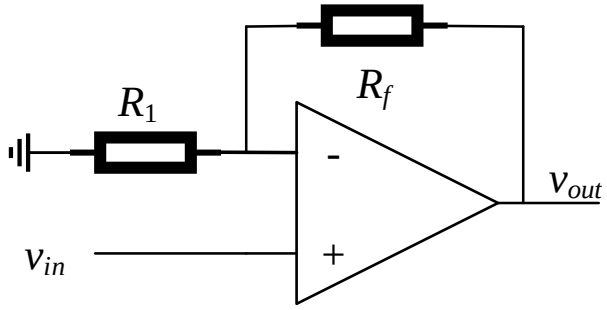


$$v_{out} = -\frac{R_f}{R_i} v_{in}$$



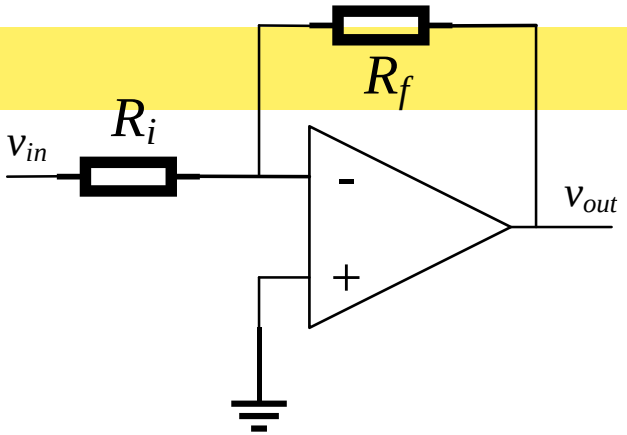
$$v_{out} = -\frac{C(x)}{C_f} v_{in}$$

Operational Amplifier - 3



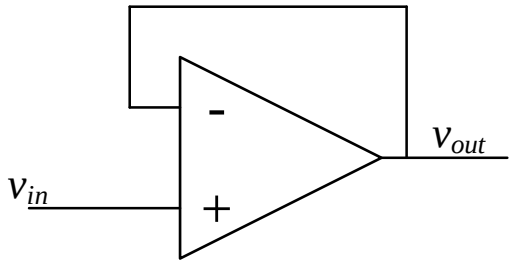
Non-inverting amplifier

$v_{out} = ?$



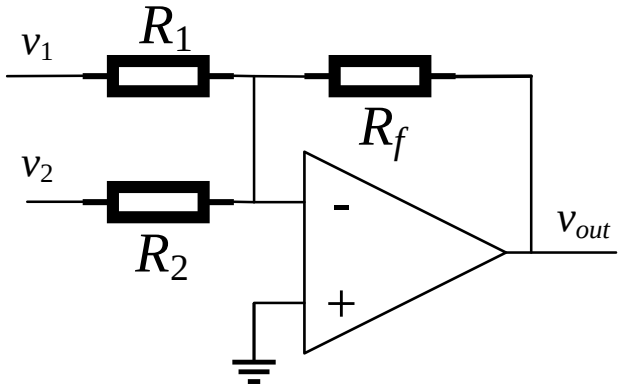
Inverting amp

$v_{out} = ?$



Follower

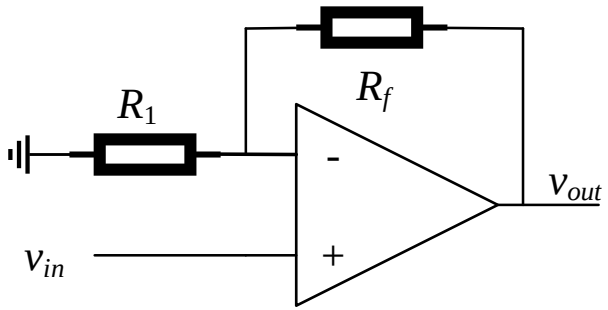
$v_{out} = ?$



Summing amp

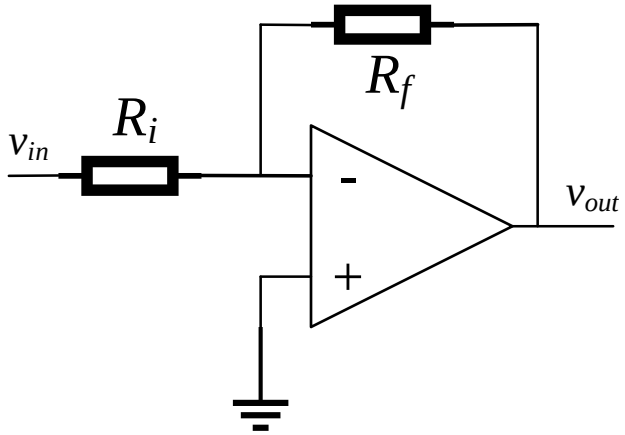
$v_{out} = ?$

Operational Amplifier - 4



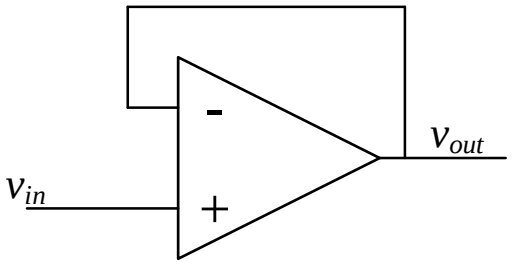
Non-inverting amplifier

$$v_{out} = \left(1 + \frac{R_f}{R_1}\right) v_{in}$$



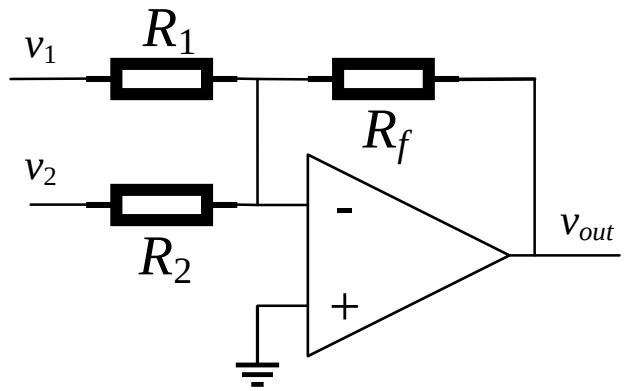
Inverting amp

$$v_{out} = -\frac{R_f}{R_i} v_{in}$$



Follower

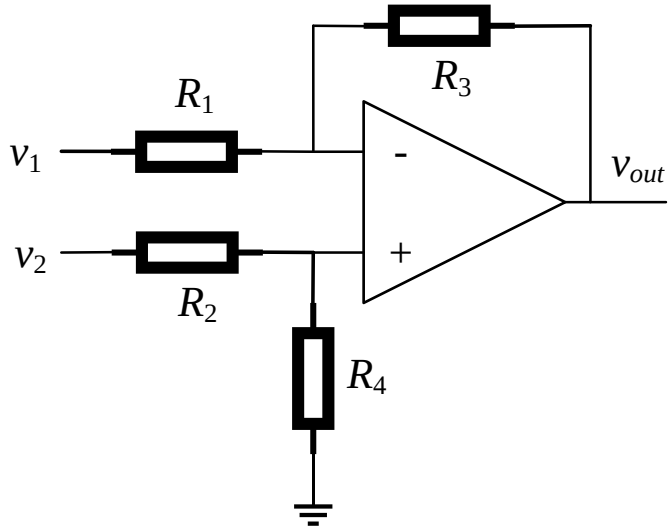
$$v_{out} = v_{in}$$



Summing amp

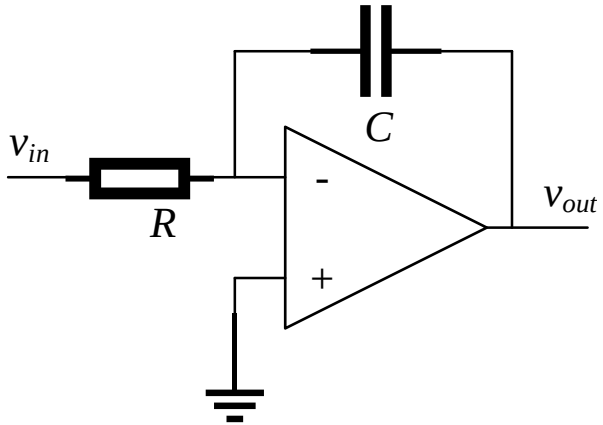
$$v_{out} = -R_f \left(\frac{v_1}{R_1} + \frac{v_2}{R_2} \right)$$

Operational Amplifier - 5



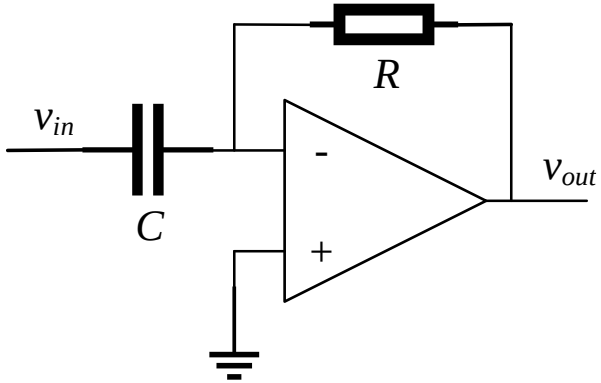
Differential amplifier

$v_{out} = ?$



Integrator

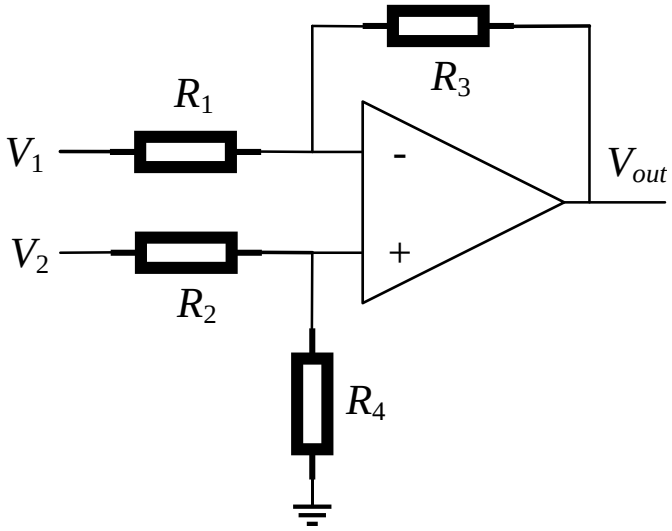
$v_{out} = ?$



Differentiator

$v_{out} = ?$

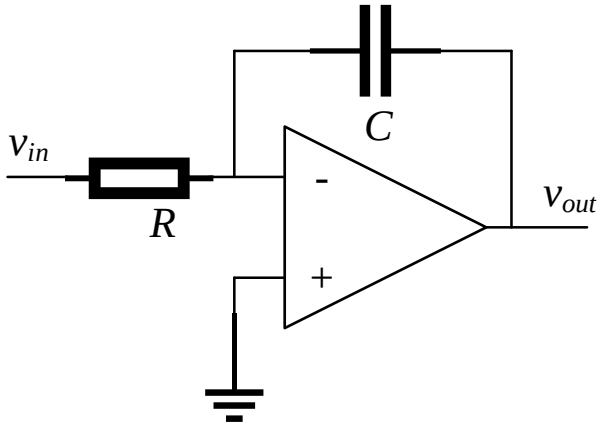
Operational Amplifier - 6



Differential amplifier

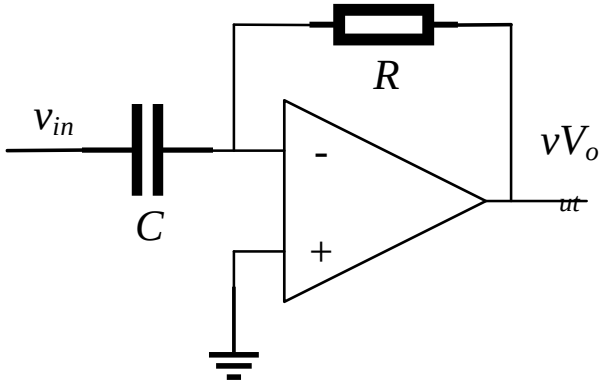
$$v_{out} = \frac{R_1 + R_3}{R_4 + R_2}\frac{R_4}{R_1}v_2 - \frac{R_3}{R_1}v_1$$

$$v_{out} = \frac{R_3}{R_1}(v_2 - v_1) \qquad \text{if} \quad R_2 = R_1, R_4 = R_3$$



Integrator

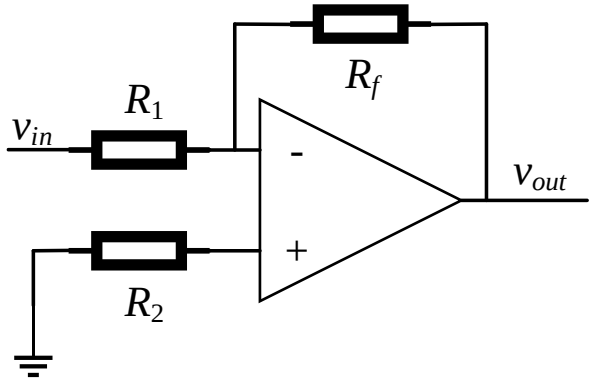
$$v_{out} = -\int_0^t \frac{v_{in}}{RC}dt$$



Differentiator

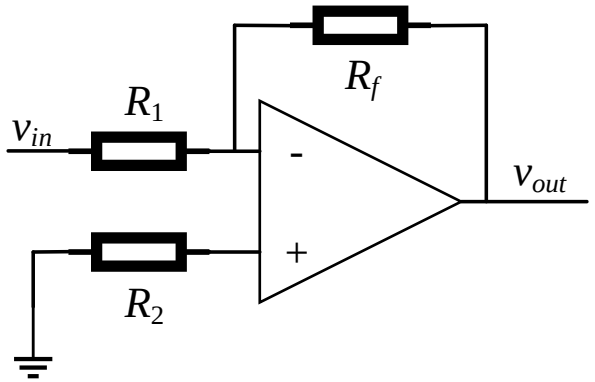
$$v_{out} = -RC\frac{dv_{in}}{dt}$$

Operational Amplifier - 7



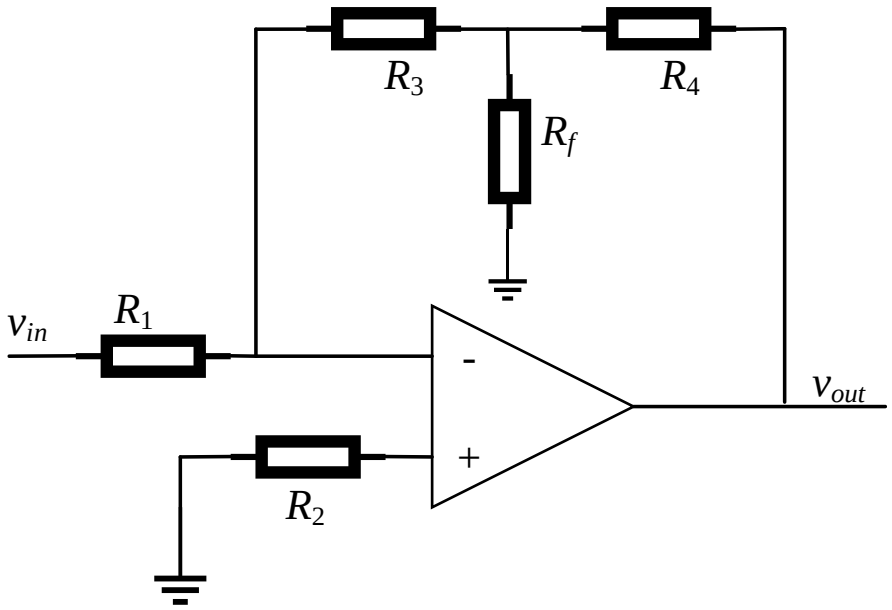
$$K = \frac{v_{out}}{v_{in}} = ?$$

Operational Amplifier - 8



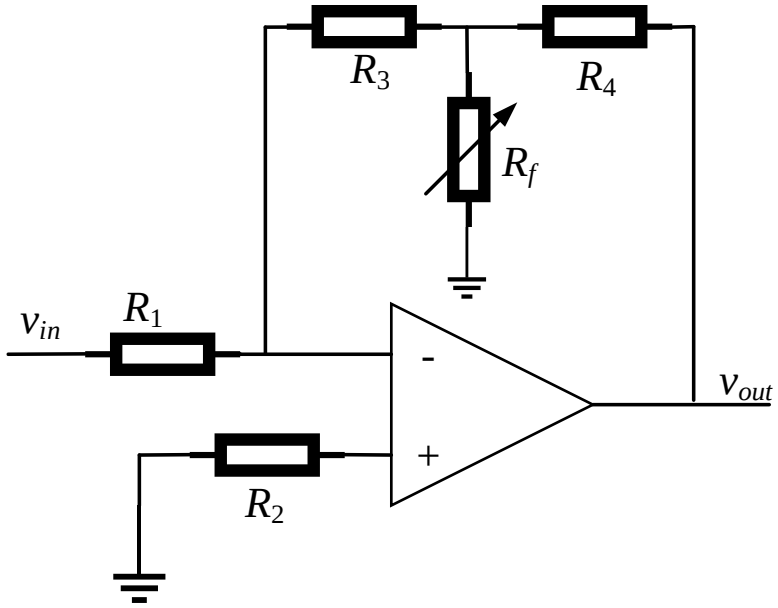
$$K = \frac{v_{out}}{v_{in}} = -\frac{R_f}{R_1}$$

An amplifier with an amplification controlled by R_f - 1



$$K = \frac{v_{out}}{v_{in}} = ?$$

An amplifier with an amplification controlled by R_f - 2



$$K = \frac{v_{out}}{v_{in}} = -\frac{R_3}{R_1}\left(1 + \frac{R_4}{R_3} + \frac{R_4}{R_f}\right)$$

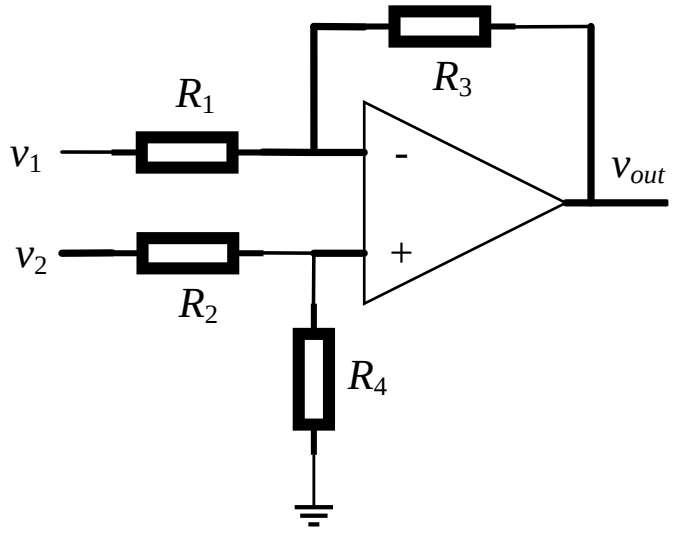
Differential amplifiers

- A diagram of the differential amplifier is,

$$v_{out} = \frac{R_1 + R_3}{R_4 + R_2}\frac{R_4}{R_1}v_2 - \frac{R_3}{R_1}v_1 \qquad v_{out} \propto (v_2 - v_1)$$

- The circuit presented is the differential amplifier if the values of resistors in the circuit satisfy the relation

$$\frac{R_1 + R_3}{R_4 + R_2}\frac{R_4}{R_1} = \frac{R_3}{R_1} \longrightarrow \frac{R_3}{R_1} = \frac{R_4}{R_2} \quad \text{or} \quad R_1R_4 = R_2R_3$$



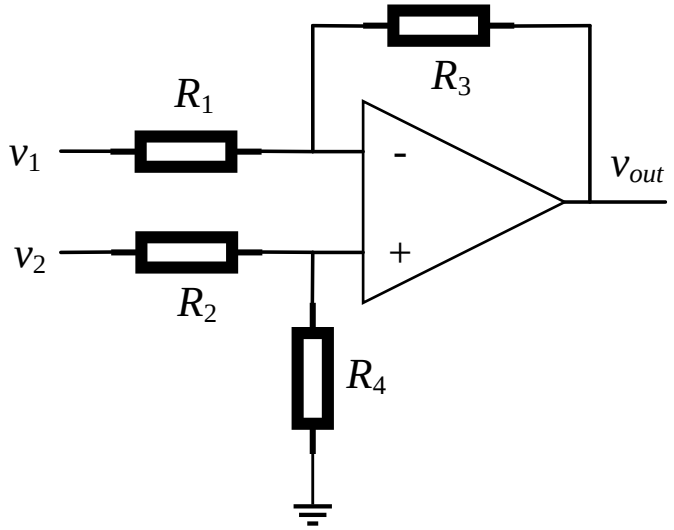
- The gain k_d of differential voltage is equal to:

$$v_{out} = \frac{R_3}{R_1}(v_2 - v_1) \longrightarrow k_d = \frac{R_3}{R_1} = \frac{R_4}{R_2}$$

Instrumentation amplifier - 1

- The control of amplification factor k_d in the circuit in the right figure is possible only by simultaneously changing the values of two resistors in the circuit, which is inconvenient. Below Equation must still be satisfied.

$$\frac{R_3}{R_1} = \frac{R_4}{R_2}$$



- There are limit resistance in the non-inverting and inverting input

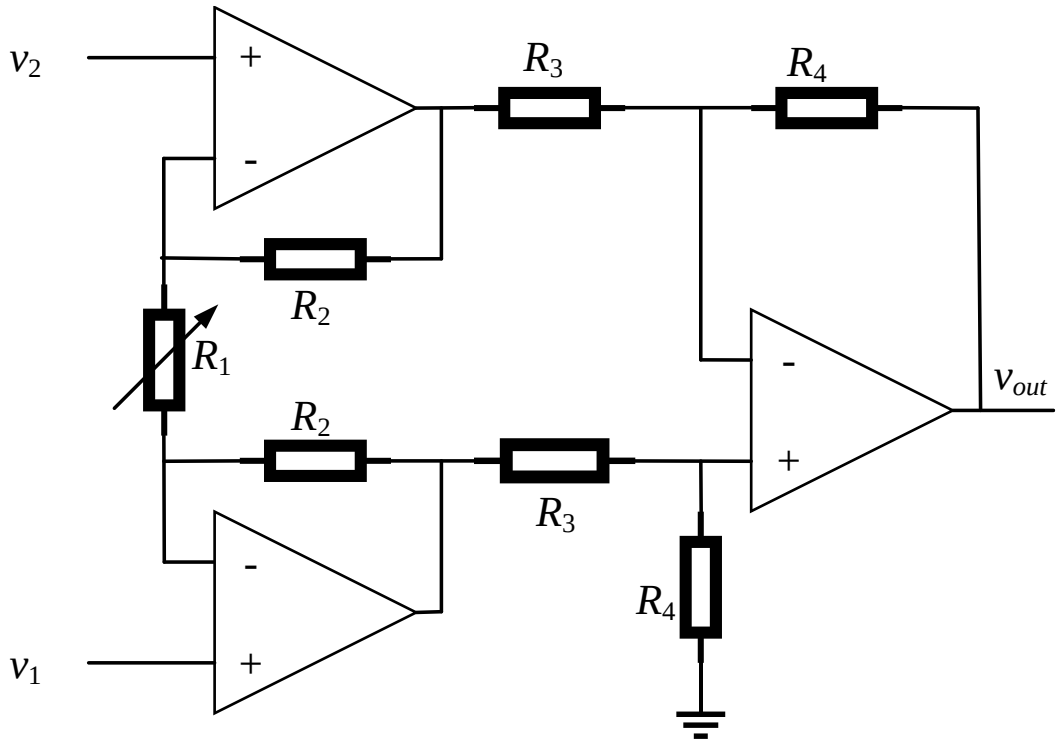
- The instrumentation amplifier is characterized by better parameters and an easier method of amplification control.

Instrumentation amplifier - 2

- The gain of the instrumentation amplifier is expressed

$$v_{out} = \left(\frac{R_4}{R_3} + \frac{2R_2R_4}{R_1\left(R_3 + R_4\right)} \right)\left(v_1 - v_2\right)$$

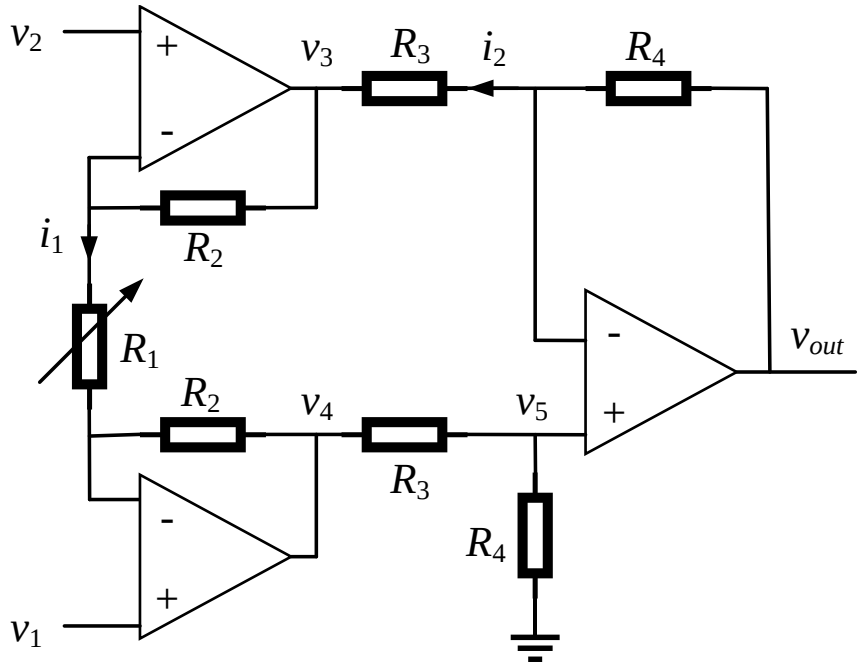
- Control of the instrumentation amplifier is realized by changing the values of resistor R_1



- This resistor can be connected as an external resistor, or it can be included in the internal structure of the integrated device

- The instrumentation amplifiers are manufactured in the form of integrated circuits.

Instrumentation amplifier - 3



$$i_1 = \frac{v_2 - v_1}{R_1}$$

$$v_3 = \frac{v_2 - v_1}{R_1} R_2 + v_2 = \frac{(R_1 + R_2)v_2 - R_2 v_1}{R_1}$$

$$v_4 = -\frac{v_2 - v_1}{R_1} R_2 + v_1 = \frac{(R_1 + R_2)v_1 - R_2 v_2}{R_1}$$

$$i_2 = \frac{v_5 - v_3}{R_3} = \frac{R_4 v_4 / (R_3 + R_4) - v_3}{R_3} = \frac{R_4 v_4 - R_3 v_3 - R_4 v_3}{R_3 (R_3 + R_4)}$$

$$= \frac{R_1 R_4 (v_1 - v_2) + R_2 R_3 (v_1 - v_2) - R_1 R_3 v_2}{R_1 R_3 (R_3 + R_4)}$$

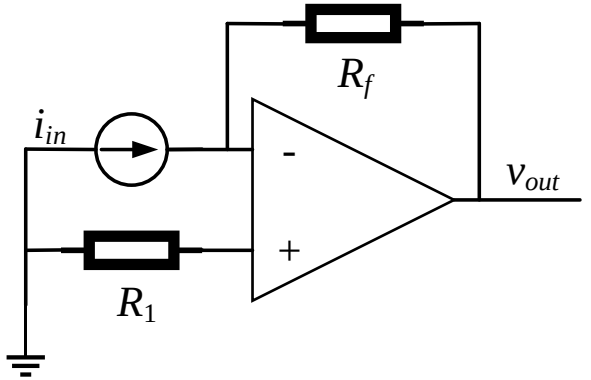
$$v_{out} = i_2 R_4 + v_5 = R_4 \frac{R_1 R_4 (v_1 - v_2) + R_2 R_3 (v_1 - v_2) - R_1 R_3 v_2}{R_1 R_3 (R_3 + R_4)} + \frac{R_4}{R_3 + R_4} \cdot \frac{(R_1 + R_2)v_1 - R_2 v_2}{R_1}$$

$$= \left(\frac{R_4}{R_3} + \frac{2R_2 R_4}{R_1 (R_3 + R_4)} \right) (v_1 - v_2)$$

Current-to-voltage converters

- Amplifiers of high current sensitivity are also used in measurement circuits and systems
- In measurement circuits, they have the function of current-to-voltage converters, as shown in below figure.
- The conversion of current value to voltage in the converter is realized according to the following

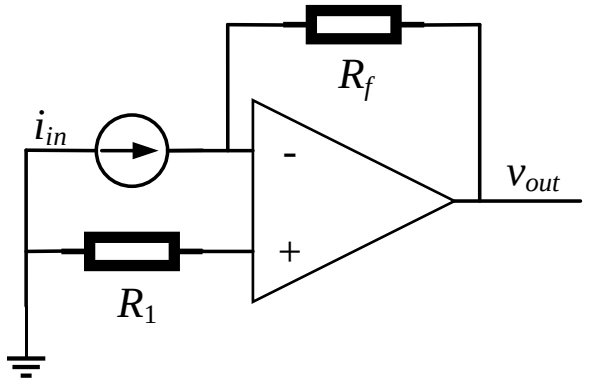
$$v_{out} = ?$$



Current-to-voltage converters

- Amplifiers of high current sensitivity are also used in measurement circuits and systems
- In measurement circuits, they have the function of current-to-voltage converters, as shown in below figure.
- The conversion of current value to voltage in the converter is realized according to the following

$$v_{out} = -R_f i_{in}$$



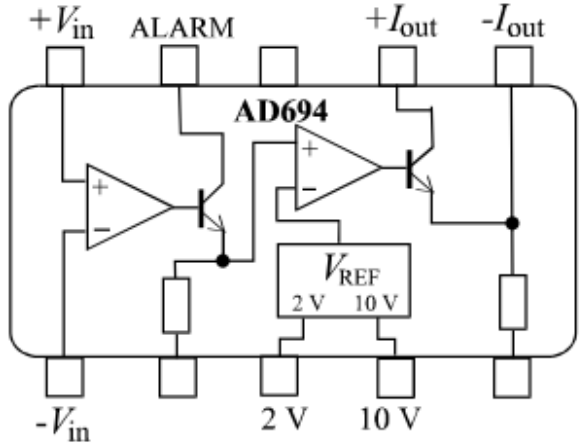
Voltage conditioners - 1

- An analog circuit matching the signal with the operational range of an ADC is called a conditioner, or a signal standardizing circuit
- Conditioners are manufactured
  - Typical ranges of output signals in conditioners are from 0V to +5V, from −5V to +5V, from 0 to 20 mA, and from 4 to 20 mA (transmitter，变送器)
- Conditioners, which are analog converters of common use, usually have their ranges of input voltage (optionally) and output current ranges defined

Voltage conditioners - 2

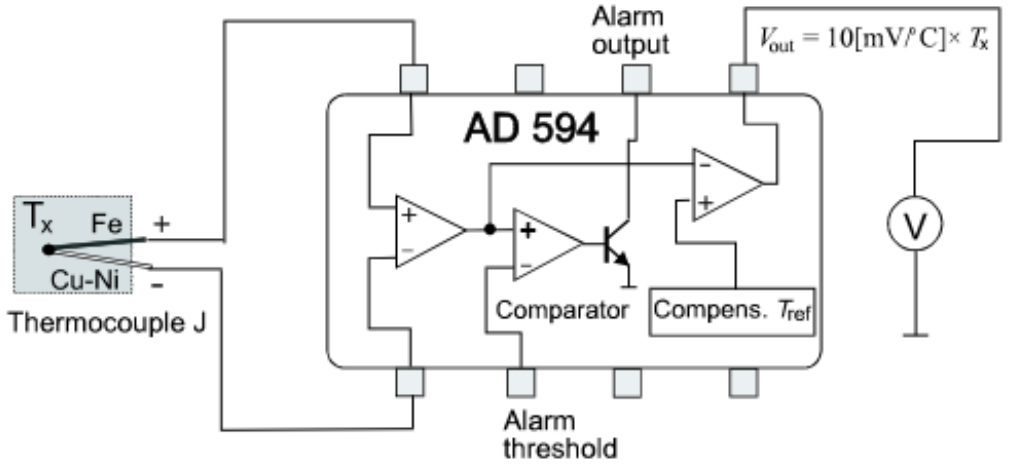
- An example of such a conditioner is the monolithic integrated circuit AD694. Its block diagram

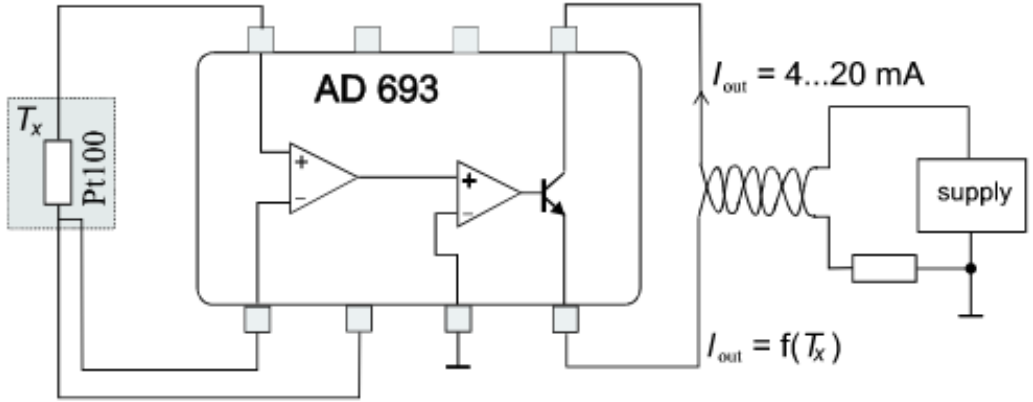
- The AD694 conditioner has two ranges of input voltage—0V to 2V and 0V to 10V—selected through keying two control terminals
- The conditioner output signal is an output current in a range of 4 to 20 mA



- The AD693 conditioner is a much more universal circuit
- The AD693 integrated circuit is adapted to cooperate with:
  - Signal source of low voltage: 0 to 30 mV or 0 to 60 mV;
  - Resistance temperature sensor (Pt100 or Ni100) in bridge;
  - Thermocouple.

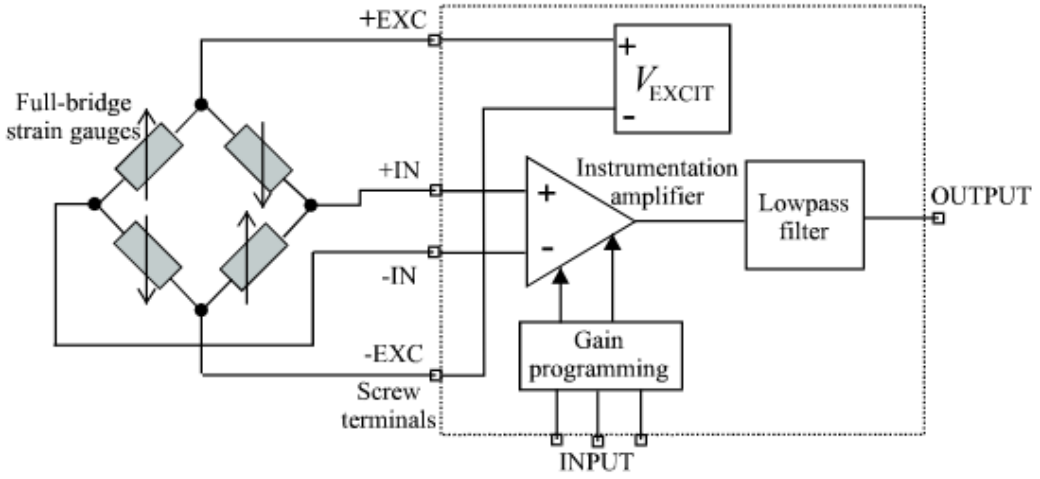
Conditioners for temperature sensors





Conditioners for strain gauges sensors

- Because signals from strain and pressure gauges are very small, the conditioning for those signals is essential
- As it is described before, resistive strain gauges operate in Wheatstone bridge in three configuration modes: quarter-bridge (with one active gauge), half-bridge (with two active gauges), and full-bridge (with four active gauges)
- The basic structure of a signal conditioner for strain gauges is,

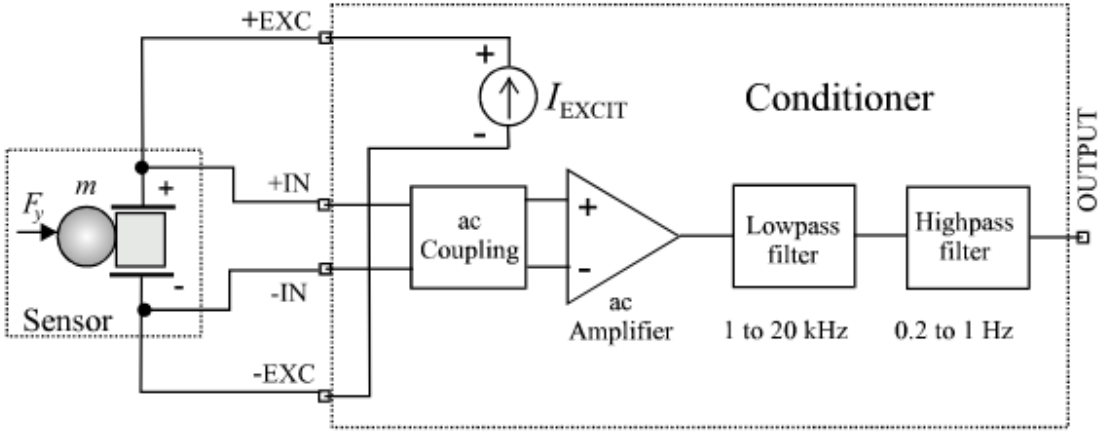


- Different conditioner devices manufactured by National Instruments are intended to operate in a different gauge configuration mode. The SCC-SG01 and the SCC-SG02 dual-channel modules

Conditioners for piezoelectric sensors

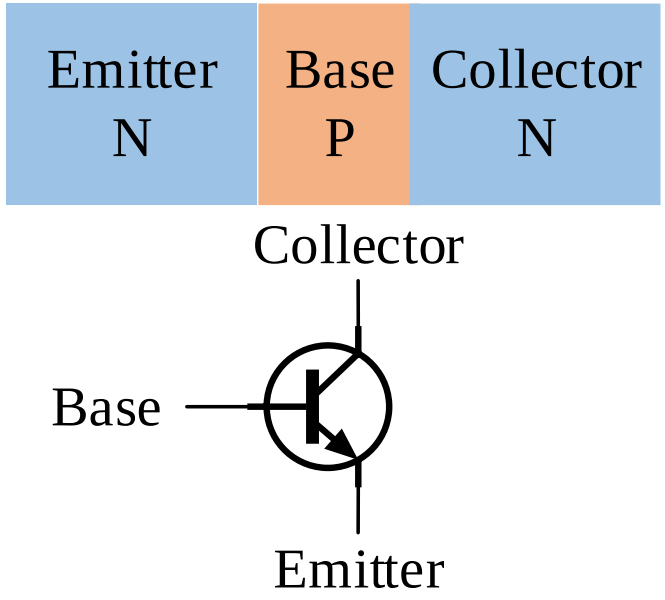
- Piezoelectric sensors need a current excitation. Their output signal is an alternating voltage
- A signal conditioner for a piezoelectric sensor includes an amplifier, a DC excitation source (typically 1 to 10 mA), a lowpass filter with the upper limit frequency up to 20 kHz, as well as a highpass filter
- The latter, with the low limit frequency from 0.1 to 1 Hz, filters a dc voltage and low-frequency components in the signal spectrum

- Multichannel conditioners also have Sample and Hold (S&H) circuits
- A signal conditioner for a piezoelectric accelerometer is ,

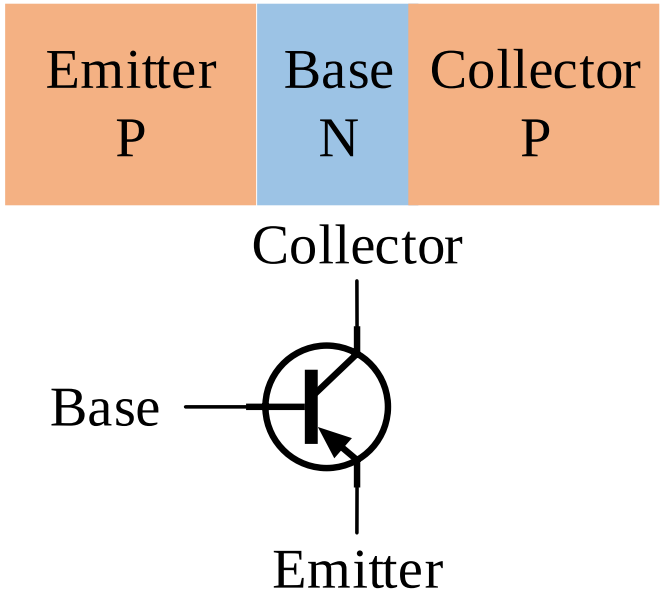


Transistors

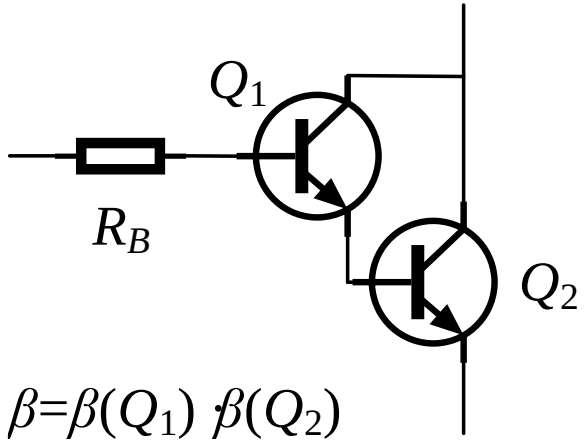
- An alternative is to use transistors (which themselves are in amplifiers)



NPN Transistor



PNP Transistor



Darlington Pair
Transistor Circuit

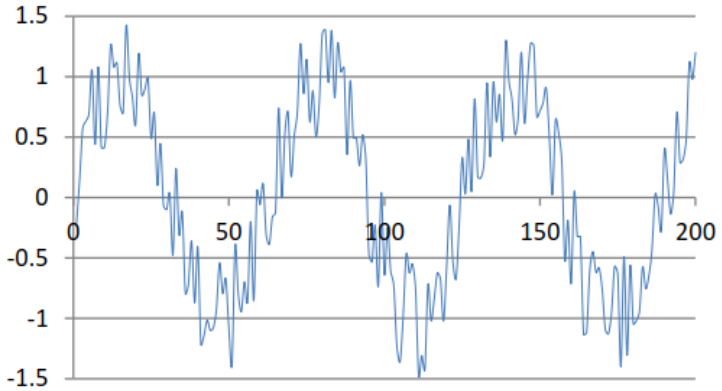
Addition of noise

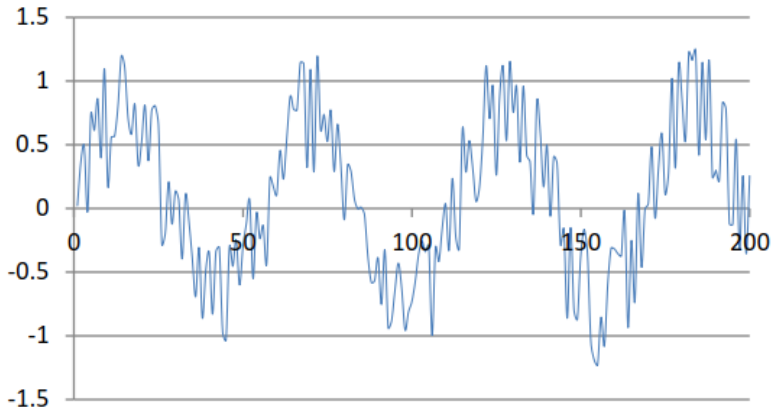
- When two independent sources of noise are added it is the mean square values which add, in contrast to two signal sources where the rms values are added i.e

Signal sources $v_{sum}=v_1+v_2+\ldots$

Noise sources $v_{sum}^2=v_1^2+v^2+\ldots$

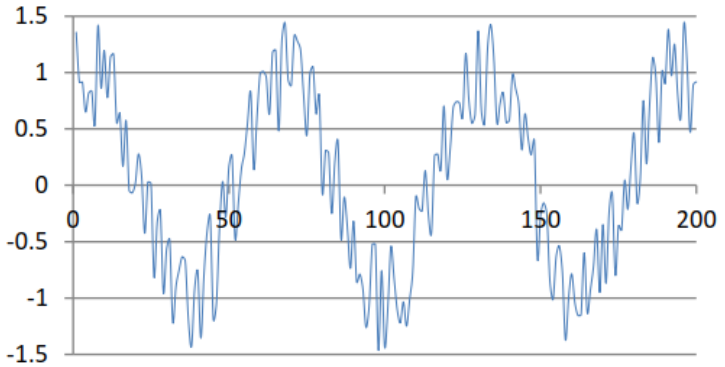
Signal 1: 1V (0.29V RMS noise)

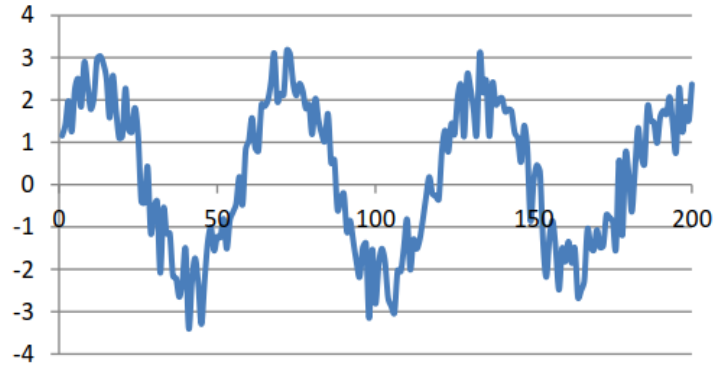




Signal 2: 0.75V (0.29V RMS noise)

Signal 3: 1V (0.29V RMS noise)

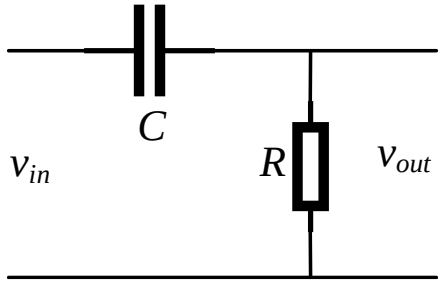




Resultant: 2.75V (0.5V RMS noise)

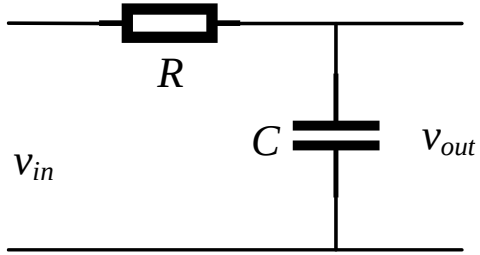
Passive filter

Passive High pass filter

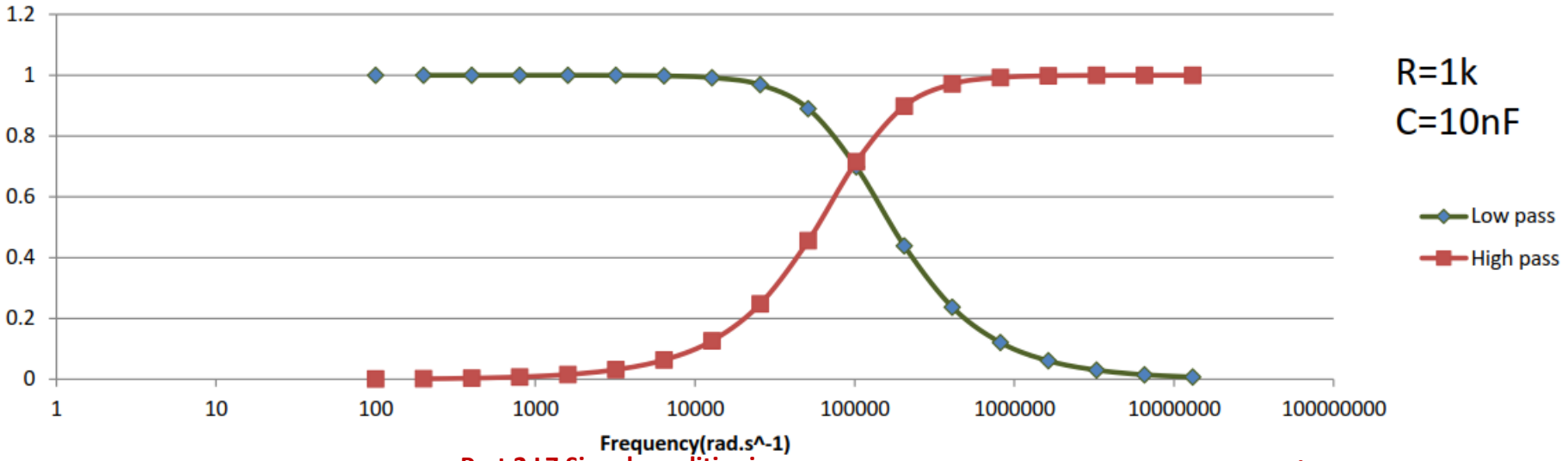


$$|v_{out}| = |v_{in}| \frac{\omega RC}{\sqrt{1+(\omega RC)^2}}$$

Passive Low pass filter

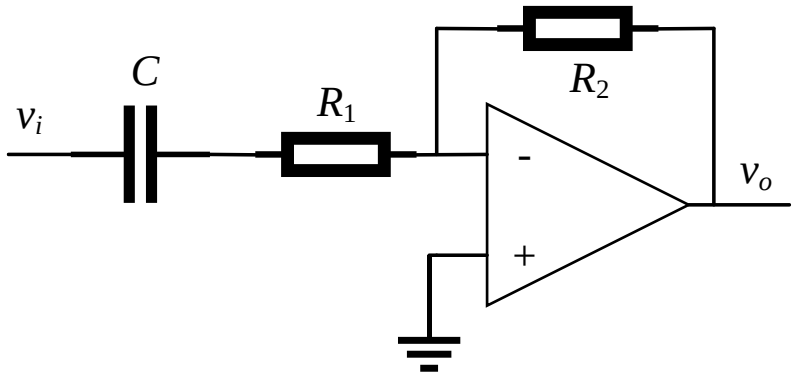


$$|v_{out}| = |v_{in}| \frac{1}{\sqrt{1+(\omega RC)^2}}$$



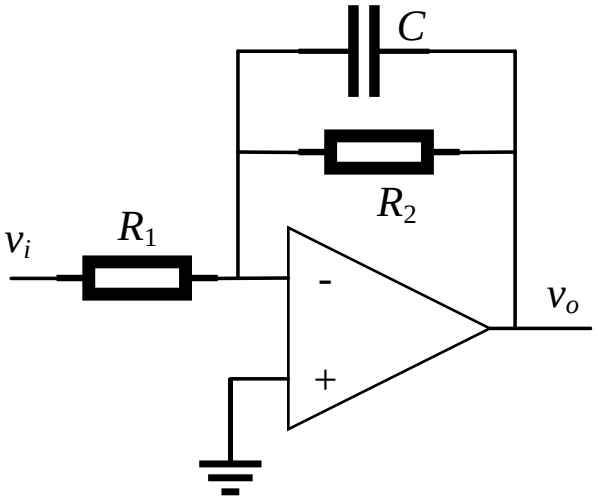
Active filter

Active High pass filter



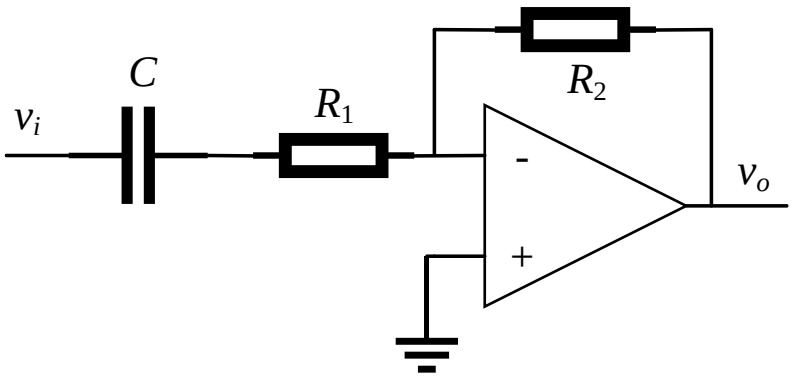
$$\frac{v_o}{v_i} = ?$$

Active Low pass filter



$$\frac{v_o}{v_i} = ?$$

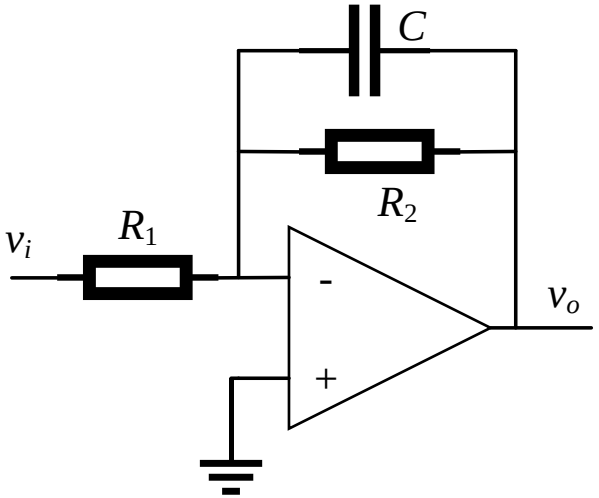
Active High pass filter



$$\frac{v_o}{v_i} = -\frac{R_2}{R_1 + 1/j\omega C}$$

$$= -\frac{j\omega R_2 C}{j\omega R_1 C + 1}$$

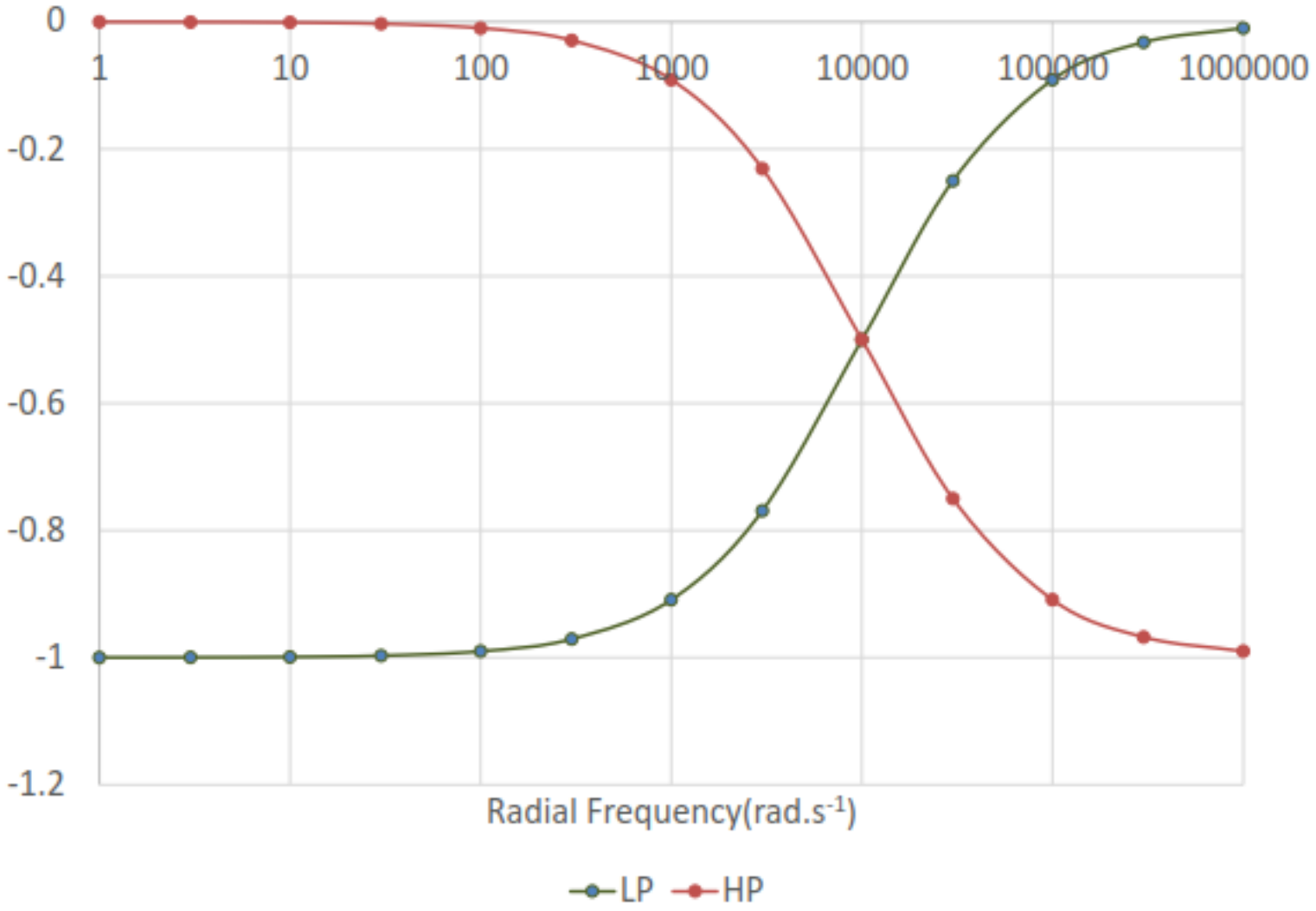
Active Low pass filter



$$\frac{V_o}{V_i} = -\frac{\frac{R_2 \, 1/j\omega C}{R_2 + 1/j\omega C}}{R_1}$$

$$= -\frac{R_2}{R_1}\frac{1}{j\omega R_2 C + 1}$$

$R_1 = 100\text{k}, R_2 = 100\text{k}, C = 1\text{nF}$



Home question

- If we have three signals which have the average signal and noise level of

Signal 1 = 100 ± 10 mV (10%)

Signal 2 = 150 ± 20 mV (13.3%)

Signal 3 = 25 ± 25 mV (100%)

What is the percentage noise on the summed signal?